Overall Score: ________

Fall 2025 Final Exam

Foundations of Data Science

Name: ____

Question:	1	2	3	4	5	6	7	Total
Points:	16	14	23	17	9	11	10	100

Instructions

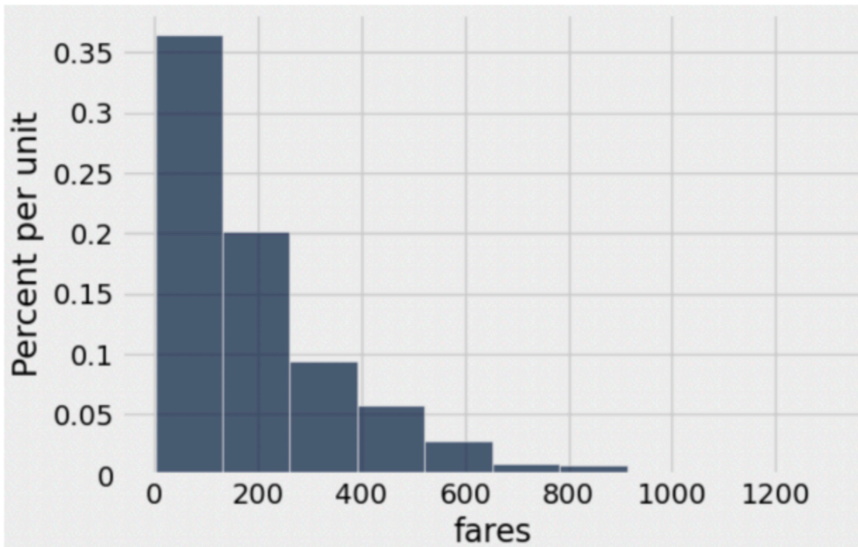
- Make sure you have a copy of the Final Exam Reference Guide and the Table Reference.
- Select the correct response(s) or provide a written response depending on the question type. If a prompt asks you to write code, then you can provide your own code or use the provided template, if available. Try to provide your responses in the template blanks or boxed spaces provided. If you find that you need additional space, write your extended response(s) on one of the provided blank sheets of paper and number them, so we can connect your response to the question.
- You can assume the following code has been run, when you are writing your Python code:

```
from datascience import *
import numpy as np
import matplotlib+
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
```

- For multiple-choice questions (◯), select only one answer.
- For multiple-answer questions (☐), select all correct options. You can earn partial credit, but incorrect choices will lower your score. However, your score will never be negative.
- The open response questions will be graded as:
  - Full Points: The response is correct and may contain a very very small error.
  - Partial Points: A reasonable response was provided. The partial point value will depend on your response.
  - No Points: No reasonable attempt was provided.
- Once you are finished, turn in your exam and you are welcome to leave.
- Please, complete the Post Survey posted on Canvas if you have not done so already.

1. **BART.** Your data analyst team is interested in studying Bay Area public transportation, so you begin analyzing data from the widely-used BART train system and the AC Transit bus services for the year 2022. Unfortunately, due to budget cuts, your available compute power is unable to process all of the data from 2022, so instead, your team is going to work with a large random sample of 1,000 riders. That sample is loaded into a table called `transport`. The first few rows of that table are previewed in the Table Reference section.

   (a) (3 points) Given below is the distribution of the fares column from the `transport` table. Which of the following conclusions can you draw from the plot? **Select all that apply.**

   √ **The distribution of the `'fares'` column in `transport` is right-skewed.**

   ☐ The distribution of the `'fares'` column in `transport` is left-skewed.

   √ **The median of the `'fares'` column in `transport` is less than the mean.**

   ☐ The median of the `'fares'` column in `transport` is greater than the mean.

   (b) (2 points) Which of the following statements must be true? **Choose one.**

   ◯ The distribution of fare spending among all riders is approximately normal.

   √ **The distribution of sample means of fare spending is approximately normal for large random samples of data.**

   ◯ The distribution of sample medians of fare spending is approximately normal for large random samples of data.

   ◯ The distribution of sample means of fare spending is skewed for large random samples.

   ◯ None of the above.

   (c) Your team is interested in estimating the proportion of all riders who had transferred between a BART train and an AC Transit bus at least once. You decide to use your sample of 1,000 riders to estimate this unknown population parameter.

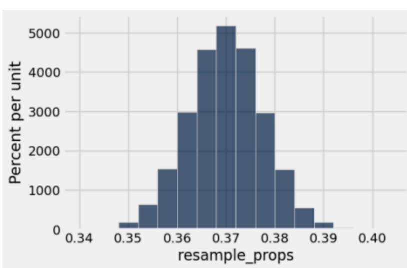
      i. (3 points) Provide code that will generate a table of 10,000 bootstrapped proportions of riders who transferred between a BART train and an AC Transit bus at least once.

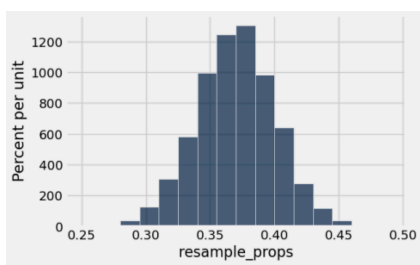
```
resample_props = make_array()

for i in np.arange(10_000):
    resamp = ____________________________________________
    resamp_prop = _______________________________________
    resample_props = ___________________________________________

resamp_props_tbl =  Table().with_column("resample_props", resample_props)
```

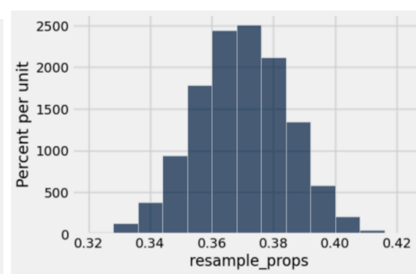
Sample Solution:

```
resample_props = make_array()

for i in np.arange(10_000):
    resamp = transport.sample()
    resamp_prop = np.mean(resamp.column("transfer"))
    resample_props = np.append(resample_props, resamp_prop)

resamp_props_tbl = Table().with_column("resample_props", resample_props)
```

ii. (2 points) You find that the mean and standard deviation of your bootstrapped proportions, resample props is 0.37 and 0.015, respectively. Which of the following most closely resembles the distribution of resample props? **Choose one.**







◯ The left graph ◯ The middle graph √ **The right graph**

iii. (2 points) Provide code that creates the array `interval` which contains the left and right endpoints of a 95% confidence interval estimate for the proportion of riders in the population who transferred at least once.

Note: You may use variable names defined from previous sub-parts in your code.

```
left = ________________________________________
right = ________________________________________
interval = make_array(left, right)
```

Sample Solution:

```
left = percentile(2.5, resample_props)
right = percentile(97.5, resample_props)
interval = make_array(left, right)
```

iv. (2 points) Which of the following conclusions can you draw using the 95% confidence interval generated in the previous part (iii)? **Choose one.**

- ◯ If someone takes the BART train, there is a 95% chance that they transfer to an AC Transit bus.
- √ **If you make confidence intervals from many large random samples from the population, you can expect that roughly 95% of the intervals you create will contain the true population proportion.**
- ◯ There is a 95% chance that the population's true transfer proportion is within the interval generated in part (iii).
- ◯ There is a 95% chance that the sample's true transfer proportion is within the interval generated in part (iii).
- ◯ None of the above.

v. (2 points) Your team has one last request. They want your 95% confidence interval to be no wider than 3%. What is the smallest sample size that satisfies this requirement? **Choose one**.

- ◯ `(2 * 0.5 / 0.03) ** 4`
- √ `(4 * 0.5 / 0.03) ** 2`
- ◯ `(2 * 0.03 / 0.5) ** 4`
- ◯ `(4 * 0.03 / 0.5) ** 2`
- ◯ None of the above.

2. **NHANES.** The National Health and Nutrition Examination Survey (NHANES) is a health survey conducted by the U.S. Centers for Disease Control and Prevention (CDC). It combines interviews, physical examinations, and laboratory tests to assess the health, nutrition, and lifestyle of people living in the United States. It is one of the most comprehensive health datasets available and is used for research, clinical standards, and public health decision-making. Data is collected every year.

The data in NHANES is organized into categories, each with its own set of variables. A small sample of the many variables and their data type are listed below.

- Numerical values: study year, age, glucose level, calories consumed, blood pressure, body mass index (BMI), hours of sleep, cholesterol level
- Categorical values: race/ethnicity, diabetes (yes/no), smoker (yes/no), heart disease (yes/no)

(a) (2 points) Using the NHANES data, the standard visualization to understand how mean BMI of US adults has changed between from 1960 to and 2020 is _____. **Choose one.**

◯ a bar chart √ **a line plot** ◯ a scatter plot ◯ a histogram

(b) (2 points) Using the NHANES data, the standard visualization to understand the distribution of race/ethnicity is _____. **Choose one.**

√ **a bar chart** ◯ a line plot ◯ a scatter plot ◯ a histogram

(c) (2 points) Using the NHANES data, the standard visualization to understand the distribution of hours of sleep per night for an adult is _____. **Choose one.**

◯ a bar chart ◯ a line plot ◯ a scatter plot √ **a histogram**

(d) (2 points) Choose the best technique to answer the question 'Do smokers in the US have a different mean blood pressure than non-smokers?'. **Choose one.**

◯ Randomized control experiment

√ **Hypothesis test**

◯ Confidence interval

◯ Linear regression

◯ Classification

(e) (2 points) Choose the best technique to answer the question 'Can we predict whether someone has diabetes based on their body mass index (BMI), age, and fasting glucose level?'. **Choose one.**

◯ Randomized control experiment

◯ Hypothesis test

◯ Confidence interval

◯ Linear regression

√ **Classification**

(f) (2 points) Choose the best technique to answer the question 'What is the mean age of people with heart disease?'. Choose one.

◯ Randomized control experiment

◯ Hypothesis test

√ **Confidence interval**

◯ Linear regression

◯ Classification

(g) (2 points) Choose the best technique to answer the question 'Can we predict a person's cholesterol level from their calories consumed?'. Choose one.

◯ Randomized control experiment

◯ Hypothesis test

◯ Confidence interval

√ **Linear regression**

◯ Classification

3. **Offside Rule.** Soccer has an *offside rule*, that designates when an attacking player is NOT allowed to touch the ball, or be involved in the play. Soccer's law-making body, the International Football Association Board (IFAB), is considering a change to the rule. The current rule and the proposed new rule are summarized below.

   - *Current rule*: A player is offside if ANY part of their body that they can play the ball with, is beyond the last defender when their teammate kicks the ball.
   - *New rule*: A player is offside if ALL parts of their body that they can play the ball with, are beyond the last defender when their teammate kicks the ball.

   IFAB believes that the new rule will benefit the attackers, and as a result, it will increase the number of goals scored in a game. They have been conducting a study over the past year, where they have implemented the *new rule* in all games in the youth leagues in Italy, Sweden, and the Netherlands. IFAB has not implemented the rule anywhere else. They will then compare this year's goals scored using the *new rule*, to last year's goals scored using the *current rule*, for the same teams.

   (a) (2 points) Which of the following is a correct null hypothesis that could be used to test whether the *new rule* increased the number of goals scored per game? **Choose one.**

   √ **There is no difference between the average number of goals scored per game with the *new rule* and the average number of goals scored per game with the *current rule.***

   ○ There is a positive difference between the average number of goals scored per game with the *new rule* and the average number of goals scored per game with the *current rule*.

   ○ There is a negative difference between the average number of goals scored per game with the *new rule* and the average number of goals scored per game with the *current rule*.

   (b) (2 points) Which of the following is a correct alternative hypothesis that could be used to test whether the *new rule* increased the number of goals scored per game? **Choose one.**

   ○ There is no difference between the average number of goals scored per game with the *new rule* and the average number of goals scored per game with the *current rule*.

   √ **There is a positive difference between the average number of goals scored per game with the *new rule* and the average number of goals scored per game with the *current rule.***

   ○ There is a negative difference between the average number of goals scored per game with the *new rule* and the average number of goals scored per game with the *current rule*.

   (c) (2 points) Suppose that IFAB stores the data for the study in a table called `soccer_games`. The first few rows of a hypothetical table are previewed in the Table Reference section. Each row is an individual game from this year or last year for the youth leagues described above. The table shows us the number of goals scored by each team in the game (`'home_goals'` and `'away_goals'`), but it doesn't tell us the total number of goals. Create a python function called `total_goals` that has 2 arguments, `home` and `away` where:

   - `home` is the number of goals the home team scored.
   - `away` is the number of goals the away team scored.

   The function should return the total number of goals scored in the game.

Sample Solution:

```
def total_goals(home, away):
    return home + away
```

(d) (2 points) Using the `total_goals` function, update the `soccer_games` table by adding a column called `'goals'` to the `soccer_games` table showing the total number of goals scored for each game.

Sample Solution:

```
soccer_games = soccer_games.with_column('goals',
    soccer_games.apply(total_goals, 'home_goals', 'away_goals')
)
```

(e) (2 points) The `'offside'` column in the updated `soccer_games` table indicates whether a game was played under the `'new rule'` or the `'current rule'`. Provide code that will generate a histogram showing the distribution of the total number of goals scored per game under the new rule overlaid with the distribution of the total number of goals scored per game under the current rule.

Sample Solution:

```
soccer_games.hist('goals', group='offside')
```

(f) (3 points) The test statistic we will use is average number of goals scored per game under the new rule minus the average number of goals scored per game under the current rule. Write code defining the function `diff_of_means` which will take a table (e.g. `soccer_games`) as argument and return the value of the test statistic. You can assume the input table has a column called `'offside'` with string values and a column called `goals` with integer values, but may have more columns.

```
def diff_of_means(tbl):
   means_table = __________________________
   means_array = __________________________
   return _________________________
```

Sample Solution:

```
def diff_of_means(tbl):
    means_table = tbl.select('offside','goals').group('offside', np.mean)
    means_array = means_table.column('goals mean')
    return means_array.item(1) - means_array.item(0)
```

(g) (2 points) Which of the following statements are true? **Choose one.**

- √ **Large positive values of the test statistic support the alternative hypothesis.**
- ◯ Large negative values of the test statistic support the null hypothesis.
- ◯ None of the above.

(h) (3 points) Next, we simulate generating data many times under the null hypothesis. Complete the code below which will run a permutation of the data and calculate the test statistic 1,000 times, and store the results in an array called `diffs_array`.

```
diffs_array = _______________________
repetitions = 1000
for _______________________________:
    resampled_table = soccer_games.______________________
    shuffled_labels = resampled_table.column('offside')
    shuffled_table = soccer_games.with_column('offside', ______________)
    shuffled_diff = diff_of_means(_________________)
    diffs_array = _________________________
```

Sample Solution:

```
diffs_array = make_array()
repetitions = 1000
for i in np.arange(repetitions):
    resampled_table = soccer_games.sample(with_replacement = False)
    shuffled_labels = resampled_table.column('offside')
    shuffled_table = soccer_games.with_column('offside', shuffled_labels)
    shuffled_diff = diff_of_means(shuffled_table)
    diffs_array = np.append(diffs_array, shuffled_diff)
```

(i) (2 points) Given the `diffs_array` computed from the 1000 simulations, and an observed test statistic named `obs_diff_avgs`, write code that will calculate the p-value for this hypothesis test.

Sample Solution:

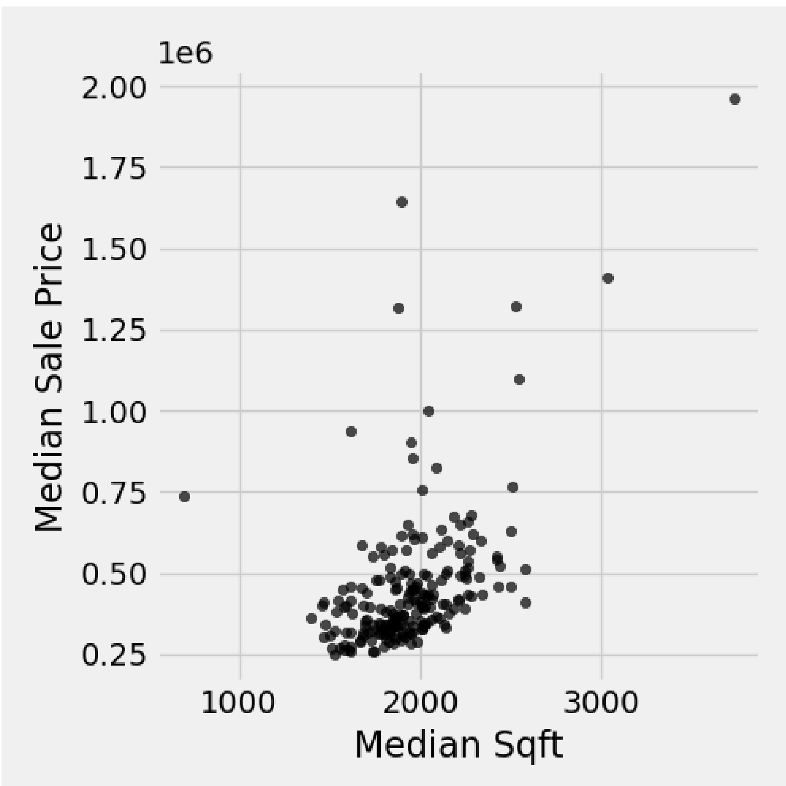
```
np.count_nonzero(diffs_array >= obs_diff_avgs) / 1000
```

(j) (3 points) If the p-value is 10%, which of the following is a valid conclusion for this hypothesis test? **Choose all that apply.**

- √ **10% of the test statistics simulated under the null hypothesis were as extreme as, or more extreme than, the observed test statistic.**
- ☐ The new rule increases the number of goals scored each game by 10%.
- √ **With a p-value cutoff of 5%, our data are consistent with the null hypothesis.**
- ☐ With a p-value cutoff of 5%, our data are consistent with the alternative hypothesis.
- ☐ There is a statistically significant increase in goals scored in games using the new rule, compared to the current rule.

4. **Housing.** The table `housing` contains data from Zillow on median sale prices and median square footage for recent new constructions in several US cities. The first few rows of that table are previewed in the Table Reference section. You and your coworker set out to study the relationship between these two variables.

(a) (2 points) Write code to create the following scatterplot of median sale price vs. median sqft.

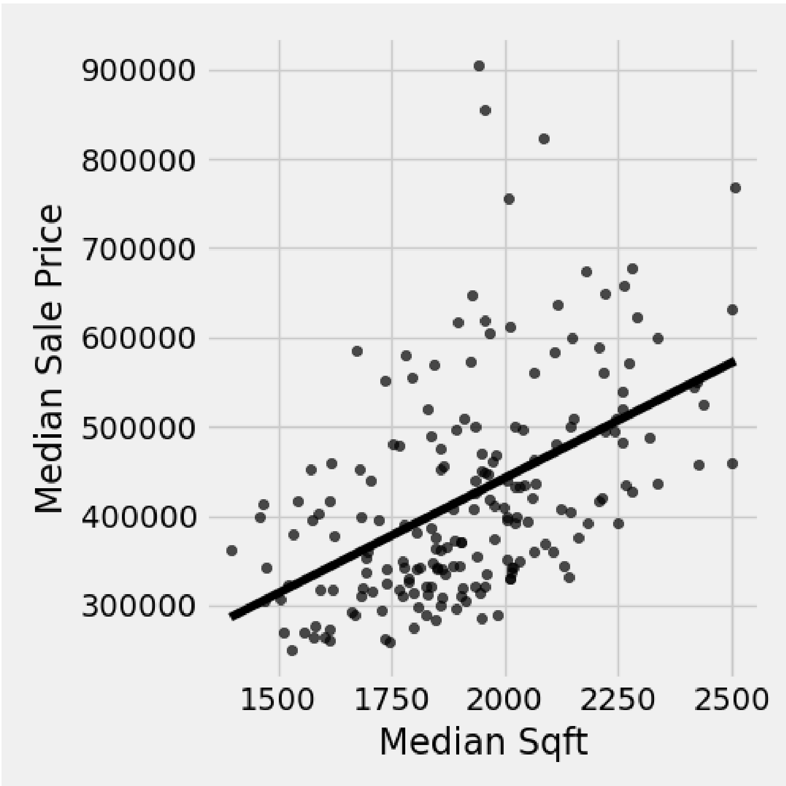


Sample Solution:

```
housing.scatter('Median Sqft', 'Median Sale Price')
```

(b) Some cities (such as San Jose and San Francisco) have unusually high values that distort the graphic. We have created the table `housing_filtered`, which contains only rows from `housing` where both `'Median Sqft'` and `'Median Sale Price'` are within 2 standard deviations of their respective means.

The scatterplot of the resulting `housing_filtered` table is shown below with the least squares regression line included.



i. (3 points) Based on the scatterplot associated with `housing_filtered`, which of the following statements are true? **Choose all that apply.**

√ **The graphic shows a positive relationship between median square footage and median sale price across cities.**

☐ The graphic shows a negative relationship between median square footage and median sale price across cities.

√ **Cities with higher median square footage tend to have higher median sale prices.**

☐ Cities with higher median square footage tend to have lower median sale prices.

☐ Houses with larger square footage tend to have higher sale prices.

☐ Houses with larger square footage tend to have lower sale prices.

(c) (3 points) You ask your coworker to compute the correlation coefficient `r` between these two variables (using `housing_filtered`), but you notice that their code has at least one mistake in it. In the code below, circle and cross out each mistake and, if applicable, write the correct code immediately above. Alternatively, you can rewrite the code in the solution box. You can assume that the following `standard_units` function from a lecture does not contain any errors:

```
def standard_units(any_numbers):
    '''Convert any array of numbers to standard units.'''
    return (any_numbers - np.mean(any_numbers))/np.std(any_numbers)
```

Here is your coworker's code:

```
median_sale_price_in_su = standard_units('Median Sale Price')
median_sqft_in_su = standard_units('Median Sqft')
r = np.sum(median_sale_price_in_su * median_sqft_in_su)
```

Sample Solution:

```
median_sale_price_in_su = standard_units(
    housing_filtered.column('Median Sale Price'))
median_sqft_in_su = standard_units(
    housing_filtered.column('Median Sqft'))
r = np.mean(median_sale_price_in_su * median_sqft_in_su )
```

(d) (3 points) You calculate the following statistics on the `housing_filtered` data:

- mean of 'Median Sqft': 1921.805 sq ft
- standard deviation of 'Median Sqft': 232.287 sq ft
- mean of 'Median Sale Price': \$422,382.825
- standard deviation of 'Median Sale Price': \$120,463.915
- correlation coefficient (r): 0.498

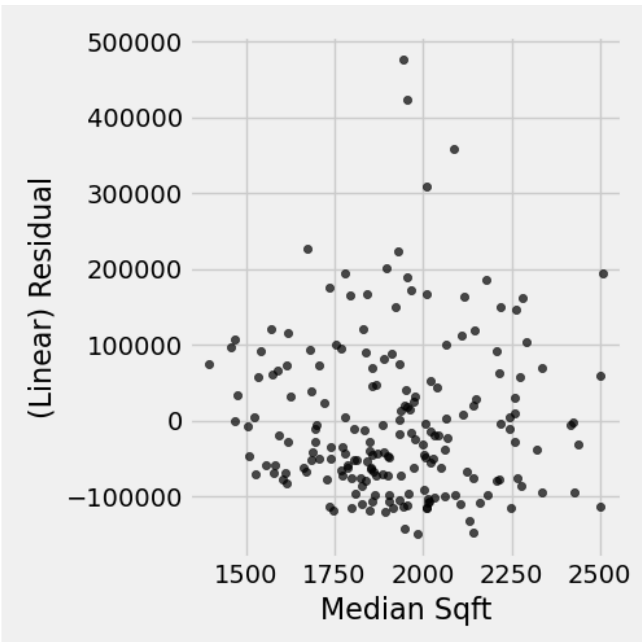
Using the numerical summaries and the linear regression line given above, answer the following. What is the predicted median sale price for a city with a median square footage of 2,300 sq ft? You should show how you compute the `slope` and `intercept` using the provided numerical values, but you do *not* need to simplify these to final numerical values.

```
slope = _________________________________________________________________
intercept =  ____________________________________________________________
predicted_median_sales_price_for_2300 = _________________________________
predicted_median_sales_price_for_2300
```

Sample Solution:

```
slope = 0.498 * 120_463.915 / 232.287
intercept = 422_382.825 - 1_921.805 * slope
predicted_median_sales_price_for_2300 = slope * 2300 + intercept
predicted_median_sales_price_for_2300
```

(e) (2 points) After reflecting on the scatter plot with fit line for the `housing_filtered` table, you think that a linear model doesn't seem like a good fit. You generate the following residual plot to confirm your suspicion:



In the solution box below, provide at least one reason why the generated residual plot confirms that a linear model is less than ideal.

Sample Solution: The residuals should appear as a random, unstructured cloud centered vertically around zero. In this residual plot, there is clear structure, with a few large positive residuals around a median square footage of about 2,000. This indicates that the linear model systematically under-predicts for those cities, suggesting that a straight line does not adequately capture the relationship.

(f) (2 points) The RMSE for the linear model is approximately 104,487.742. You decide to fit a quadratic model $ax^2 + bx + c$ to the data. To do this, you create the following function to calculate RMSE for a quadratic model:

```
def quad_rmse(a, b, c):
```

```
    x = housing_filtered.column('Median Sqft')
    y = housing_filtered.column('Median Sale Price')
    quad_estimates = a * x ** 2 + b * x + c
    quad_residuals = y - quad_estimates
    return np.sqrt(np.mean(quad_residuals ** 2))
```

Write code to find the best fitting quadratic model for the `housing_filtered` data. Specifically, the code should produce the values for `a`, `b`, and `c`.

Sample Solution:

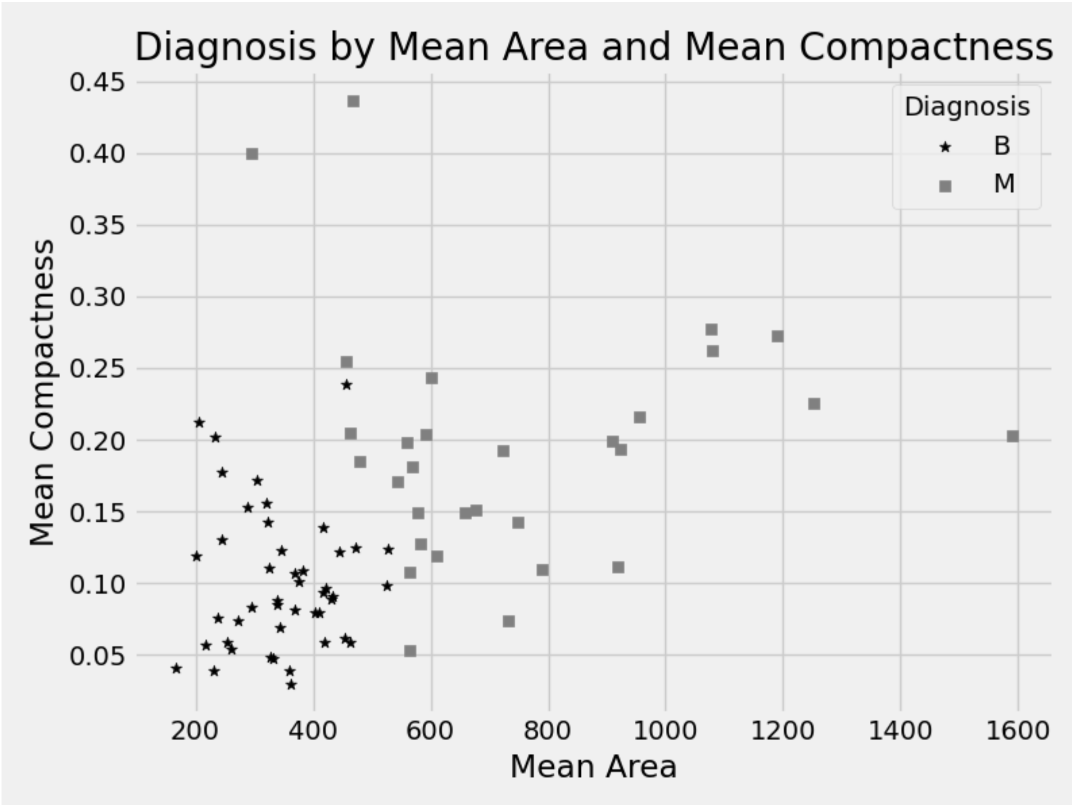
```
a, b, c = minimize(quad_rmse)
a, b, c
```

(g) (2 points) It turns out that the quadratic model fits the data a little bit better than the linear model, in terms of RMSE. What does this mean?**Choose one.**

- √ **The RMSE for the quadratic model is less than 104,487.742.**
- ○ The RMSE for the quadratic model is equal to 104,487.742.
- ○ The RMSE for the quadratic model is more than 104,487.742.

5. **Breast Cancer.** Researchers at the University of Wisconsin-Madison contributed data to support the diagnosis and prognosis of breast cancer. The data in `cell_images` contain characteristics of the cell nuclei presented in digitized images of a fine needle aspirate (FNA) of a breast mass. The first few rows of that table are previewed in the Table Reference section. We've created a kNN classifier using the features of `'Mean Area'` and `'Mean Compactness'` to predict a diagnosis of benign (`'B'`) or malignant (`'M'`).

(a) (2 points) Using the data in the following scatterplot, what would a k-nearest neighbors classifier with $k = 3$ label a cell if the mean area was 400 and the mean compactness was 0.2?



Select one.

◯ 'B' √ 'M'

(b) (2 points) Your partner on this project thinks that the data should be standardized before building the k-nearest neighbors classifier. Which of the following statements are true? **Choose one.**

◯ It doesn't matter if the data is standardized, since the set of nearest neighbors will be unchanged.

√ **Standardizing the data has an impact on the results, since the mangnitude of the features affects how distance is calculated.**

◯ None of the above.

(c) (3 points) The following functions and their descriptions are all used to build a k-NN classifier. In the space provided, indicate the order that the functions must be run (from first to last), using the numbers next to the functions.

1. `row_distance(row1, row2)`: Return the distance between two numerical rows of a table.
2. `distances(tbl, example)`: Compute distance between example row and every row in a table. Return table augmented with `'Distance'` column.
3. `majority_class(tbl)`: Return the class with the highest count in the table.
4. `closest(tbl, example, k)`: Return a table of the k closest neighbors to example
5. `distance(pt1, pt2)`: Return the distance between two points, represented as arrays.
6. `classify(table, example, k)`: Return the majority class among the k nearest neighbors of example.

Sample Solution: 5, 1, 2, 4, 3, 6

(d) (2 points) Suppose that in our training data set there are 250 benign data points and 149 malignant data points. What is the smallest value of k, that would guarantee that a k-NN classification will always classify a data point as benign?

Sample Solution: 299 (one more than double the smaller value).

6. **Final Project.**

(a) (2 points) In the Final Project you built a k Nearest Neighbors (k-NN) classifier to predict a movie's genre (comedy vs. thriller) based on characteristics of the dialogue. In the original table `movies`, which was used to create the `train`, `validation`, and `test` sets, all of the movies had a value for the genre. But sometimes data sets will have missing values. In order to build the k-NN classifier that uses the whole training set, you need to know the class value of some, but not all of the training examples. **Choose one.**

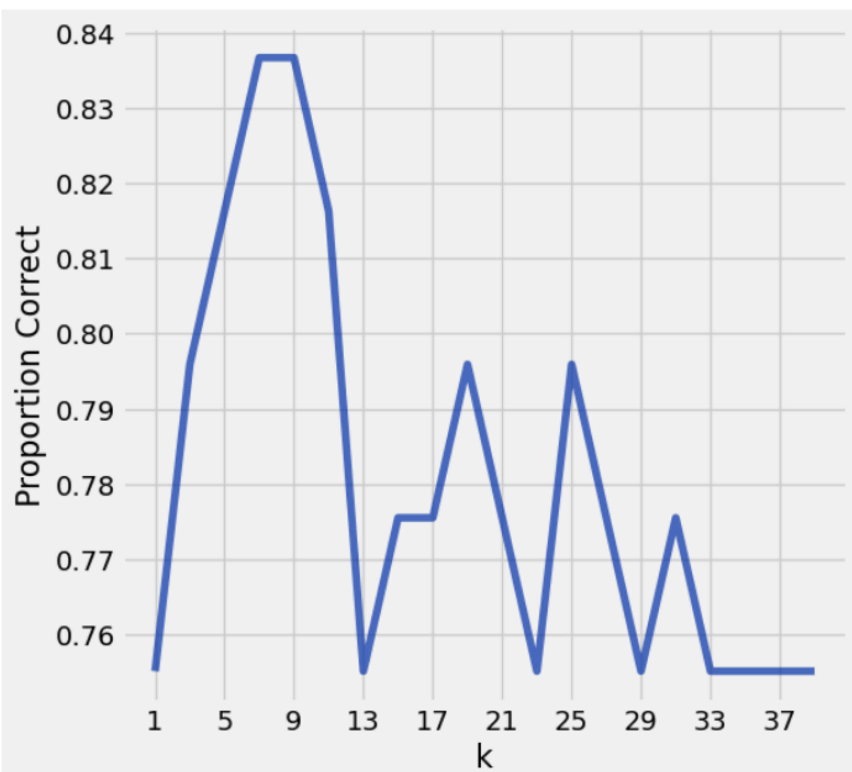
◯ True √ **False**

(b) (3 points) Fifteen percent of the original data was reserved for testing the k-nearest neighbors classifier. That data was stored in a table `test`, which is just a random sample of rows from the original `movies` table.

Assume you have a function defined that classifies an example based on a 3-nearest neighbors classifier using this data, describe in 1-3 sentences how you would calculate the accuracy of that k-nearest neighbors classifier.

Sample Solution: I would use the k-nearest neighbors classifier function to predict the label for each row in the `test` table (by finding the k closest rows from the training data and taking the majority label). Then, I would compare each predicted label to the actual label in `test` and calculate the accuracy as the proportion of correct predictions out of all rows in `test`.

(c) (3 points) As one of the last steps of the project in "Section 4.1: Fine Tuning", you produced the following line plot ranging over many odd k-values. In the space provided, write one or two sentences explaining what the graph shows us. Specifically indicate what "Proportion correct" refers to, and what important conclusion we can make from the graph.



Sample Solution: We are ”fine-tuning” the classifier. Using the validation set, we calculated the accuracy of the classifier function using many different odd k-values. We see from the graph that k-values of 7 and 9 give the highest accuracy on the validation set.

(d) (3 points) In the project you built the following functions, written in a random order (see c for descriptions).

- `row_distance(row1, row2)`, `distances(tbl, example)`, `majority_class(tbl)`, `closest(tbl, example, k)`, `distance(pt1, pt2)`

Write code for the function `classify(training, example, k)` which inputs a table of movies, an example movie, and an odd integer k, and returns the majority genre (`'comedy'` or `'thriller'`) among the k nearest neighbors fo the example. You are allowed to use any of the functions listed above.

Sample Solution:

```
def classify(training, example, k):
    closestk = closest(training, example, k)
    topkclasses = closestk.select('Class')
    return majority(topkclasses)
```

7. **Ice Cream.** Your two favorite ice cream shops, Game of Cones and Sundae Night Live, start selling mystery pints. Each mystery pint has a random flavor, and the shops put up the following signs with the possible flavors and the probability of getting each one:

Game of Cones:	
Flavor	Probability
Vanilla	0.30
Maple	0.25
Raspberry	0.45

Sundae Night Live:	
Flavor	Probability
Vanilla	0.65
Maple	0.25
Raspberry	0.10

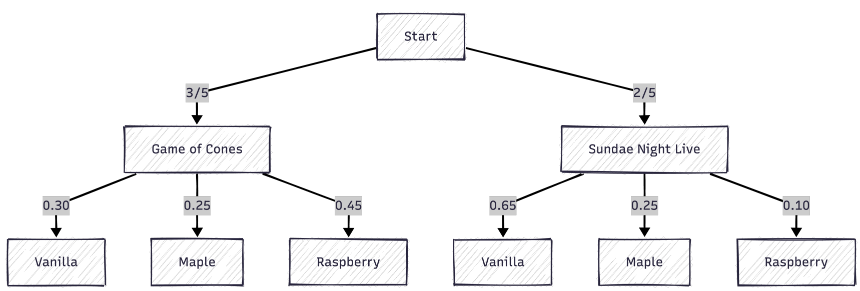
Answer the following questions. For each one, write your answer as a Python expression or an unsimplified expression—you do not need to evaluate it to get a number. If there is not enough information provided to answer the question, write "not enough information".

(a) (2 points) You buy five pints from Game of Cones. What is the probability that they are all the same flavor?

Sample Solution: $0.30^5 + 0.25^5 + 0.45^5$

(b) (2 points) An ice cream researcher decides that for an entire semester, they'll buy one pint every weekday (Mon-Fri). Because of their schedule, they go to Game of Cones three days a week and Sundae Night Live two days a week. Draw a probability tree to model their random selection on any given night.

Sample Solution:



(c) (2 points) The ice cream researcher also happens to be your roommate, and randomly picks one of the nights, in accordance with the description in part (b), and buys you a pint. If the pint was vanilla flavored, what is the probability they went to Game of Cones?

Sample Solution: $\dfrac{0.60 * 0.30}{0.60 * 0.30 + 0.40 * 0.65}$

Suppose the researcher suspects that mystery pints from Game of Cones do not follow the distribution that the shop claims (in the table above). They decide to keep track of the flavors they got from visiting Game of Thrones and then conduct a hypothesis test using the following null and alternative hypotheses:

- Null hypothesis: Mystery ice cream pints from Game of Cones are distributed according to the probability distribution specified by the shop.
- Alternative hypothesis: Mystery ice cream pints from Game of Cones are not distributed according to the probability distribution specified by the shop.

The researcher decides to use a p-value cutoff of 0.01 for their test.

(d) (2 points) What is an appropriate choice of test statistic for this test?

Sample Solution: Total variation distance between the shop's distribution of flavors and the observed distribution of flavors.

(e) (2 points) If the null hypothesis is true, what is the probability that their test will fail to reject the null hypothesis?

Sample Solution: 99%

Table Reference

transport

Here is a preview of the `transport` table and some information about the data values:

- `'id'` (**integer**): identification (id) of the rider.
- `'transfer'` (**boolean**): whether that particular rider transferred between a BART train and an AC Transit bus at least once during 2022.
- `'fares'` (**float**): total amount that particular rider spent on fares in 2022, measured in dollars.

id	transfer	fares
32849	True	12.5
29490	False	62
81305	False	131.75
70654	False	43

... (996 rows omitted)

soccer_games

Here is a preview of `soccer_games`, a hypothetical table of soccer games, and some information about the data values.

- `'home'`, `'away'` (**string**): representing the home and away team in each game.
- `'home_goals'`, `'away_goals'` (**integer**): representing the number of goals the team home or away team scored in the game.
- `'offside'` (**string**): whether the game used the `'new rule'` or `'current rule'`.

home	home_goals	away	away_goals	offside
Frosinone	3	Inter	3	new rule
Leece	2	Lazio	0	new rule
Napoli	1	Roma	1	new rule
Atalanta	2	Sassuaola	1	new rule
Monza	2	Genoa	5	new rule

housing

Here is a preview of the `housing` table and some information about the data values:

- `'RegionName'` (**string**): representing the name of the region.
- `'Median Sale Price'` (**integer**): representing the median sale price in the region.
- `'Median Sqft'` (**float**): representing the median area of homes in the region in square feet.

RegionName	Median Sale Price	Median Sqft
Abilene, TX	269900	1510.1
Albany, NY	649000	2219.32
Albuquerque, NM	399990	1683.42

... (201 rows omitted)

cell_images

Here is a preview of the `cell_images` table and some information about the data values:

- `'ID'` (**integer**): identification of the patient.
- `'Diagnosis'` (**string**): representing a benign or malignant cells.
- `'mean ...'` (**float**): several different measurements of the cells.

ID	Diagnosis	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave_points	mean symmetry	mean fractal_dimension
842302	M	14.8217	9.53843	105.33	1057.8	0.0956663	0.330747	0.355243	0.14279	0.24401	0.0679343
842517	M	15.3678	13.9713	98.366	1118.69	0.071255	0.0927733	0.1157	0.0898567	0.156697	0.0497407
84300903	M	14.6685	15.8556	95.695	1002.01	0.0867167	0.208153	0.228707	0.130493	0.1969	0.0507137
84348301	M	8.94187	16.012	59.965	327.01	0.12047	0.40826	0.328303	0.127123	0.32771	0.093216
84358402	M	14.5291	10.5971	97.5793	988.813	0.0830633	0.120803	0.218293	0.0952167	0.144953	0.0469083
843786	M	9.41817	13.4467	62.729	415.297	0.104803	0.242783	0.24334	0.0887867	0.209617	0.0685373
844359	M	13.8589	16.1377	91.9933	899.97	0.081048	0.126807	0.171213	0.09253	0.166463	0.0477597
84458202	M	10.4512	16.7823	68.2187	508.62	0.0977017	0.187663	0.12878	0.0766433	0.184687	0.0650073
844981	M	9.59877	17.8507	65.3687	427.807	0.10111	0.256107	0.253477	0.10393	0.23141	0.061613
84501001	M	9.28253	22.1063	61.2197	403.747	0.103683	0.45659	0.46991	0.106917	0.219163	0.100003

... (559 rows omitted)